CS6843 Program Analysis
Assignment 1: Introduction to LLVM

Deadline: August 10, 2025 Marks: 5

Problem Statement

The objective of this assignment is to gain hands-on experience with the LLVM infrastructure by installing it and exploring its tools.

a. Compiling to LLVM IR and Using Inbuilt LLVM Passes

- Install LLVM 20.1.7 using the installation instructions given in this section.
- Write a simple C program that has function calls and loops. Use LLVM tools to do the following:
  - Generate the (human readable) LLVM IR for the program.
  - Generate the Control Flow Graph of the functions in the program as a graph in GraphViz Dot format. View the dot graph file using xdot.
  - Use the inbuilt LLVM pass to print the callgraph of the program.
  - Use the inbuilt LLVM pass to print information about loops in the program.
- Write a PDF report with:
  - The sample C program
  - The commands used for performing each of the above tasks
  - Pass outputs and generated graphs.
  - The generated LLVM IR.

Submission Guidelines

- Submit only the PDF report named as `<YOUR_ROLLNO>.pdf`.

LLVM Installation Instructions

Setting Up the Docker Image

- Install Docker on your system. Refer to the official installation instructions.
- Download the prebuilt Docker image:
  - For **x86 systems**, download it here.
  - For **macOS (ARM/M1/M2)**, download it here.
- Run the following commands to extract and load the image:

```
# For x86 systems only (ARM users can skip this step)
gzip -d llvm20-x86.tar.gz

# Load the Docker image (choose the appropriate file)
docker load -i llvm20-x86.tar     # For x86
docker load -i llvm20-arm.tar     # For ARM (macOS)
```

- Verify the image is loaded using:

```
docker image ls
```

- If you encounter any issues, you can build the Docker image manually using the provided Dockerfile:

```
docker build -t llvm20 .
```

Running the Docker Image

- Run the container using:

```
docker run -it llvm20 /bin/bash
```

- (Optional) Use volume mount to share files:

```
docker run -v /home/username/data:/root/data -it llvm20 /bin/bash
```

Alternative Installation

You may also build LLVM directly from source (without Docker) by following the official instructions:
https://llvm.org/docs/GettingStarted.html

- Ensure you build version 20.1.7 by checking out the `llvmorg-20.1.7` git branch.

LLVM Resources

- LLVM API Reference: https://llvm.org/doxygen/index.html
- LLVM IR Reference: https://llvm.org/docs/LangRef.html

Useful Hints

Listing Available Passes

To list all passes available in your LLVM build:

```
opt --print-passes
```

This shows all passes supported by your version of ‘opt‘, including analysis and transformation passes.

Running a Pass on an LLVM IR File

You can apply a pass to an LLVM IR (.ll) file using the ‘opt‘ tool:

```
opt -passes='name-of-pass' input.ll
```